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Relevance of a Neurophysiological Marker of Attention Allocation for Children's Learning-Related Behaviors and Academic Performance

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Learning-related behaviors are important for school success. Socioeconomic disadvantage confers risk for less adaptive learning-related behaviors at school entry, yet substantial variability in school readiness exists within socioeconomically disadvantaged populations. Investigation of neurophysiological systems associated with learning-related behaviors in high-risk populations could illuminate resilience processes. This study examined the relevance of a neurophysiological measure of controlled attention allocation, amplitude of the P3b event-related potential, for learning-related behaviors and academic performance in a sample of socioeconomically disadvantaged kindergarteners. The sample consisted of 239 children from an urban, low-income community, approximately half of whom exhibited behavior problems at school entry (45% aggressive/oppositional; 64% male; 69% African American, 21% Hispanic). Results revealed that higher P3b amplitudes to target stimuli in a go/no-go task were associated with more adaptive learning-related behaviors in kindergarten. Furthermore, children's learning-related behaviors in kindergarten mediated a positive indirect effect of P3b amplitude on growth in academic performance from kindergarten to 1st grade. Given that P3b amplitude reflects attention allocation processes, these findings build on the scientific justification for interventions targeting young children's attention skills in order to promote effective learning-related behaviors and academic achievement within socioeconomically disadvantaged populations.

Keywords: event-related potentials, selective attention, learning-related behaviors, academic performance

Learning-related behaviors at school entry, such as paying attention, working independently, and persisting at challenging tasks, facilitate successful school adjustment and optimal academic achievement (Howse, Calkins, Anastopoulos, Keane, & Shelton, 2003; McClelland et al., 2007; Stipek, Newton, & Chudgar, 2010). Chronic developmental stressors, such as poverty, have been shown to impede the development of these skills, such that children growing up in high risk contexts are disproportionately behind in school readiness (Fiscella & Kitzman, 2009; McLoyd, 1998; Morgan, Farkas, Hillemeier, & Maczuga, 2009). Yet, substantial variability in school readiness exists within socioeconomically disadvantaged populations. While a growing research base has documented the social and cognitive–behavioral factors associated with positive developmental outcomes despite adversity (Masten, 2001), less is known about the neurophysiological correlates of such resilient functioning. A deeper understanding of the neurophysiological processes underlying variability in learning-related behaviors within socioeconomically disadvantaged populations could inform efforts to promote school readiness within these populations, for example by identifying key neurobehavioral systems to target for intervention (Bradshaw, Goldweber, Fishbein, & Greenberg, 2012).

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Learning-Related Behaviors and School Success

Children who exhibit positive learning-related behaviors at school entry can respond effectively to learning and task demands in the classroom, as reflected in the capacity to sit still and listen to their teacher for an extended period of time without becoming distracted and to persist in a cognitively challenging activity without becoming excessively frustrated (McClelland, Acock, & Morrison, 2006). Children with deficits in such learning-related behaviors are likely to experience difficulties in the transition to school and throughout their academic career. Indeed, numerous studies have revealed that measures of learning-related behaviors are associated with literacy and mathematics achievement in the early school years (Howse et al., 2003; McClelland, Morrison, & Holmes, 2000; Stipek et al., 2010). Furthermore, deficits in learning-related behaviors in kindergarten predict lower rates of growth in literacy and math skills between kindergarten and second grade, resulting in an achievement gap that persists through the sixth grade (McClelland et al., 2006). Thus, individual differences in learning-related behaviors at school entry are an important predictor of later academic performance.

Attention and School Success

Selective attention allocation involves the enhancement of attention to goal-relevant or otherwise salient information and the simultaneous inhibition of attention to goal-irrelevant or distracting information (Dux & Marois, 2009; Neill & Westberry, 1987). Recent studies using behavioral and observational measures of attention have suggested that such selective attention plays a particularly important role in academic achievement. For example, a meta-analysis of six large-scale datasets revealed that attention at school entry was a stronger predictor of later literacy and math achievement than were externalizing behavior problems or social skills at school entry (G. J. Duncan et al., 2007). Attention control has also been found to predict both literacy and math achievement in preschoolers above and beyond the effects of inhibitory control and working memory (Lan, Legare, Ponitz, Li, & Morrison, 2011) and to mediate the association between preschool social-emotional competence and first-grade academic achievement (Rhoades, Warren, Domitrovich, & Greenberg, 2011). Illustrating the importance of attention for long-term academic outcomes, difficulties with attention regulation and impulse control in kindergarten or earlier have been shown to predict academic underachievement into early adulthood even when sociodemographic factors are controlled (McClelland, Acock, Piccinin, Rhea, & Stallings, 2013; Vitaro, Brendgen, Larose, & Tremblay, 2005).

Attention skills may influence academic performance in part by enabling positive learning-related behaviors. For example, children with stronger selective attention skills may exhibit better attentional focus during teacher instruction and spend more time on-task during independent work. Indeed, in classroom observation studies, lower time-on-task and higher distractibility have been shown to differentiate children with learning disabilities from their normal-achieving peers (McKinney & Feagans, 1983). Children rated by their teachers as being attentive in kindergarten have also been shown to exhibit more positive learning-related behaviors in the classroom in first through sixth grade (Pagani, Fitzpatrick, & Parent, 2012). Selective attention has also been hypothesized to play a role in the effective regulation of behaviors and emotions in response to emotionally challenging situations (Rueda, Checa, & Rothbart, 2010; Rueda, Posner, & Rothbart, 2011), suggesting that children with poor selective attention skills may be more easily derailed from learning-related behavior by poorly regulated emotions.

Learning-Related Behaviors and Attention Within Disadvantaged Populations

Socioeconomically disadvantaged children are more likely to display problematic learning-related behaviors from an early age (Morgan et al., 2009), leaving them underprepared to meet the behavioral and learning demands of school (Miech, Essex, & Goldsmith, 2001). Recently, researchers have suggested that these delays in school readiness may reflect the negative impact of exposure to the stressors associated with socioeconomic disadvantage on children's neurocognitive development (Noble, Norman, & Farah, 2005; Noble, Tottenham, & Casey, 2005). In particular, neurophysiological studies have revealed differences in selective attention across socioeconomic gradients (D'Angiulli, Herdman, Stapells, & Hertzman, 2008; Hackman & Farah, 2009; Stevens, Lauinger, & Neville, 2009). However, it is clear that many children within high-risk contexts are able to engage successfully in educational settings and demonstrate developmentally appropriate growth in academic skills. Children who are academically successful in this context likely show better neurocognitive function compared with their peers, which may facilitate their success. Importantly, individual differences in neurocognitive function generally reflect a combination of genetic and environmental influences and appear to be sensitive to intervention (Posner & Rothbart, 2005).

A Neurophysiological Index of Attention Allocation: The P3b Event-Related Potential

Given the importance of attention for children's positive engagement in the classroom and academic performance, investigation of the core neurocognitive processes underlying individual differences in attention skills could provide greater insight into the neurobehavioral mechanisms that contribute to learning-related behaviors and thereby reveal potentially useful targets for intervention in at-risk children. One neurophysiological measure that is closely associated with controlled attention allocation is the P3b event-related potential. The P3b is a late positive peak in the electroencephalographic (EEG) waveform that is typically maximal at centroparietal electrodes anywhere from 300 to 900 ms following the presentation of a salient, attended visual or auditory stimulus (C. C. Duncan et al., 2009; Linden, 2005; Polich, 2007). The P3b is observed following target stimuli in tasks that require the participant to discriminate between different classes of stimuli, such as the oddball, continuous performance, and go/no-go paradigms (C. C. Duncan et al., 2009). The amplitude of the P3b is greater following stimuli to which participants are instructed to actively attend (Donchin & Cohen, 1967; Hillyard, Hink, Schwent, & Picton, 1973) and is diminished as increasingly challenging secondary tasks are added, suggesting that it reflects the amount of attentional resources, in a limited-capacity attentional system, that are available to a particular task (Donchin, 1981; Isreal, Chesney, Wickens, & Donchin, 1980). The P3b has also been implicated in working memory updating processes that occur in conjunction with conscious attention allocation (Polich, 2007). This component should be distinguished from the related P3a peak, which may be observed following highly unexpected, novel distracter stimuli and which exhibits a more fronto-central topographic distribution on the scalp and slightly earlier latency (Polich, 2007). Similarly, nontarget trials to which the participant must actively inhibit the impulse to respond, such as no-go trials in a go/no-go task, elicit an ERP peak that is topographically and functionally similar to the P3a; this peak is often referred to as the *no-go P3* (Polich, 2007). In all paradigms yielding a P3, the event elicits increased attention allocation, whether because the paradigm requires a special response, updating memory, inhibiting a prepotent response, or simply coding new information for future use. In the remainder of this paper, we will focus specifically on the P3b event-related potential as measured at centroparietal sites following target stimuli in discrimination tasks.

At a neurophysiological level, the P3b peak in the EEG waveform appears to reflect synchronous activity in multiple neocortical regions, including in particular the temporoparietal junction as well as more dorsal posterior areas of the parietal cortex (Linden, 2005), which have been implicated in attention control processes in brain imaging studies (Corbetta & Shulman, 2002). These synchronous activations have been proposed to arise from phasic activity of the locus coeruleus-norepinephrine system, which facilitates selective attention by enhancing neuronal responsivity immediately following the detection of a motivationally salient stimulus (Aston-Jones, Rajkowski, & Cohen, 1999; Nieuwenhuis, Aston-Jones, & Cohen, 2005). Low P3b amplitudes may be linked to high *tonic* and low *phasic* levels of noradrenergic signaling, a pattern which is associated with attentional distractibility and behavioral impulsivity (Aston-Jones et al., 1999; Nieuwenhuis et al., 2005).

P3b Amplitude and Psychopathology, Cognitive Functioning, and Academic Achievement

The information-processing correlates and probable neural generators of the P3b event-related potential point toward its utility as a neurophysiological measure of controlled attention allocation processes that are important for learning-related behaviors and academic performance. However, the majority of studies of individual differences in P3b amplitude have focused on associations with psychiatric diagnoses in youth and adults. Research implicates low P3b amplitudes in risk for an array of externalizing disorders including ADHD, conduct disorder, and alcohol and other substance dependence disorders (Barry, Johnstone, & Clarke, 2003; Bauer & Hesselbrock, 1999; Iacono & Malone, 2011; Patrick et al., 2006). Notably, the association of attenuated P3b amplitude with externalizing disorders was observed when the P3b was measured following target stimuli across a wide variety of tasks, including both auditory and visual oddball tasks, selective attention tasks, and continuous performance tasks (Barry et al., 2003; C. C. Duncan et al., 2009). Patrick and colleagues (2006) proposed that low P3 amplitude is an indicator of a "broad neurobiological vulnerability" underlying the high comorbidity among various externalizing disorders. However, the neurocognitive and behavioral mechanisms of this vulnerability are not yet well understood. Low P3b amplitude could reflect the contribution of impairments in attention allocation processes to the developmental risk for externalizing disorders. Because most studies have been conducted using clinical populations, less is known about the cognitive and behavioral correlates of P3b amplitude in nonclinical samples.

Those studies that have examined associations between P3b amplitude and cognitive measures in nonclinical populations have done so using samples of adolescents and adults. These studies have revealed that higher P3b amplitudes to oddball targets are associated with better attentional focusing and lower impulsivity/distractibility (Boucher et al., 2010; Määttä et al., 2005; Portin et al., 2000; Russo, De Pascalis, Varriale, & Barratt, 2008). Additionally, Harmon-Jones, Barratt, and Wigg (1997) observed that adolescents with higher P3b amplitudes to targets in a continuous performance task had lower self-rated attentional impulsivity (i.e., less difficulty maintaining focused attention) and lower "nonplanning impulsivity" (i.e., a greater tendency to plan for the future). In a sample of middle-aged construction workers, Portin and colleagues (2000) found that higher P3b amplitudes to targets in an auditory oddball task correlated with faster RTs on other tasks requiring sustained attention, whereas P3b amplitude did not correlate with RTs on more cognitively complex tasks or with performance on working memory tasks, suggesting a specific link between P3b amplitude and attention. Interestingly, Määttä et al. (2005) observed that distractible adolescents exhibited both lower average P3b amplitudes to oddball targets and a greater decrease in oddball P3b amplitudes during the last block of the task, potentially reflecting a greater decline over time in attentional resource allocation to the task.

Several studies have also revealed positive associations between P3b amplitude to oddball targets and measures of general intelligence in adults (Beauchamp & Stelmack, 2006; De Pascalis, Varriale, & Matteoli, 2008; Russo et al., 2008; Wronka, Kaiser, & Coenen, 2013), and between P3b amplitude to cue stimuli in the cued continuous performance task and intellectual giftedness in young adolescents (Liu, Xiao, Shi, & Zhao, 2011). Russo and colleagues (2008) observed that lower P3b amplitude mediated the negative association between impulsivity and IQ, suggesting that highly impulsive individuals may have performed more poorly on intelligence tests due to difficulty inhibiting attentional processing of task-irrelevant information.

Given the association of low P3b amplitudes with greater impulsivity/distractibility and poorer performance on intelligence tests among adolescents and adults, it stands to reason that P3b amplitude may also have implications for academic performance in school-age children. However, to our knowledge only two studies to date have examined associations between P3b amplitude and measures of academic performance. Both of these studies provided evidence for a positive association between P3b amplitude and concurrent reading achievement among preadolescent and early adolescent youth (Harmon-Jones et al., 1997; Hillman et al., 2012). Specifically, Harmon-Jones et al. (1997) studied a sample of 34 ethnically diverse youth between the ages of 11 and 17 and found that higher P3b amplitudes to oddball targets were associated with better performance on tests of oral reading accuracy and reading comprehension. In a larger sample of 105 ethnically diverse preadolescent children, Hillman and colleagues (2012) found that higher P3b amplitudes to target stimuli in a go/no-go task were associated with better performance on a standardized reading test. Furthermore, they found that this association held above and beyond the effects of socioeconomic status, grade level, and IQ. These studies leave many intriguing research questions unanswered, such as what cognitive or psychosocial processes underlie the association between P3b amplitude and academic performance and at what point in a child's school experience this association emerges.

The Present Study

There is no published research on the relevance of P3b amplitude for learning-related behaviors and academic performance in the early elementary years in at-risk populations. The present study addressed this gap utilizing a community-based sample of primarily Black and Hispanic children living in low-income urban neighborhoods. Within this at-risk sample, we examined associations between kindergarten children's P3b amplitudes to target stimuli in a go/no-go task, teacher ratings of their learning engagement in the kindergarten classroom, independent observer ratings of their task engagement during a kindergarten cognitive assessment, and teacher ratings of their academic performance in kindergarten and first grade. We hypothesized that children with higher P3b amplitudes would exhibit more positive learning-related behaviors according to both teacher and observer ratings and would have higher teacher-rated academic performance. Furthermore, we hypothesized that P3b amplitude would indirectly influence children's growth in academic performance through its association with more positive learning-related behaviors in the classroom. Finally, we examined the degree to which P3b amplitude uniquely predicted children's learning-related behaviors above and beyond the variance accounted for by more easily collected behavioral assessments of cognitive functioning.

Method

Participants

The sample was drawn from the PATHS to Success study, which included a randomized controlled trial of a multicomponent intervention targeting kindergarten children with early-onset aggressive/oppositional behaviors, as well as a developmental study of a comparison group of children with relatively low levels of aggressive/oppositional behaviors. All children were recruited from elementary schools within a single urban school district in central Pennsylvania. At the beginning of the school year in 2008 (Cohort I) and 2009 (Cohort II), all kindergarten teachers completed a 10-item aggressive/oppositional behavior rating scale for each child in their class. More details regarding recruitment can be found in Gatzke-Kopp, Greenberg, Fortunato, and Coccia (2012). Briefly, for purposes of the intervention, 207 children in the upper quartile of aggressive behavior within each classroom were enrolled, 100 of whom were randomly assigned to the intervention condition. Since the present analyses are not concerned with intervention effects, the 100 children assigned to the intervention condition were excluded from the analysis sample. Additionally, 132 comparison children matched on sex and classroom with children in the high-aggression group were selected from those children who scored in the lowest quartile on the aggression screening measure. The analysis sample thus consists of 239 children, 44.8% of whom were in the high-aggression group at study entry. Parents provided informed consent, and children were asked to give verbal assent for all study procedures. The study procedures were approved by the Pennsylvania State University Internal Review Board.

Sixty-four percent of the children in the analysis sample were male. Consistent with the demographics of the region, 69% of the children were African American, 21% were Hispanic or Latino, 9% were Caucasian, and 1% were Asian. The average age at initial screening was 5.7 years (*SD* = .36, range = 5.1–7.0). Sample children lived in a community with very low socioeconomic resources; 79% of students in the district were classified as low-income (qualifying for free or reduced-price school meals), the majority of households were headed by a single mother (69%), and 79% of parents were estimated to have no more than a high school education. Regional statistics indicate that property crimes were twice as high and violent crimes were 4.5 times as high as comparable statistics for the entire state.

Measures

The measures that were used in the present analyses are described below. Univariate statistics for these measures are provided in Table 1.

Teacher ratings. Teachers were asked to complete questionnaires regarding each participating child in the winter of the child's kindergarten school year and in the spring of the child's first-grade year. Teachers who returned their questionnaires received a $15 gift card as compensation for their time. These questionnaires included questions about children's behavior in the classroom, relations with peers, academic performance, quality of the student–teacher relationship, and parental involvement in school. For the present analyses, kindergarten teachers' reports of children's engagement in the classroom learning environment and kindergarten and first-grade teachers' reports of children's academic performance are used. These measures are described below.

Learning engagement. Teachers rated children on 14 items describing their ability to engage productively in the classroom learning environment. This measure was adapted from a scale utilized by Bierman et al. (2008). Representative items included "This child has the self-control necessary to do well in school," "This child is careful with his or her work," and "This child can get along well enough with other children to succeed in school." Responses were provided on a 6-point Likert scale ranging from 1 (*strongly disagree*) to 6 (*strongly agree*). Although this measure was administered in all 3 years, only the kindergarten scores are used in the present analyses. Summary scores were calculated as the mean of the item scores (α = .97). Learning engagement scores were missing due to kindergarten teacher nonresponse for 28 children, or 12% of the analysis sample.

Academic performance. Teachers rated children's academic performance in the areas of reading and preliteracy skills, math skills, and overall academic functioning relative to classroom expectations on a scale of 1 (*near the very bottom of your class*) to 5 (*near the very top of your class*), using a scale developed for the Head Start REDI Project (http://headstartredi.ssri.psu.edu/). Summary scores were calculated as the mean across these three items. For the present analyses, kindergarten and first-grade academic performance scores were used. These mean scores exhibited very high internal reliability in the current sample (kindergarten: α = .99; first grade: α = .96). Academic performance measures were missing due to teacher nonresponse for 32 children in kindergarten (13% of the analysis sample) and 45 children in first grade (19% of the analysis sample).

Table 1
Univariate Statistics for All Analysis Variables

	N	*M*	*SD*	Observed range	Possible range	Skewness	Kurtosis
Go/No-Go task							
P3b mean amplitude (μV)	211	3.4	2.3	−2.7–10.0	NA	0.42	0.16
Mean response time (ms)	217	509.4	55.8	331.9–680.3	100–1,000	0.11	0.29
Go error rate (%)	217	31.3	15.1	1.8–76.9	0–100	0.54	−0.27
No-Go error rate (%)	217	38.6	15.4	2.9–96.2	0–100	0.81	1.70
Cognitive assessment							
Expressive vocabulary	236	83.9	12.3	55.0–122.0	NA	0.22	−0.13
Inhibitory control (original scale)	231	14.0	3.2	0.0–16.0	0–16	−2.21	4.82
Inhibitory control (binary)	231	0.6	0.5	0.0–1.0	0–1	NA	NA
Attentional set shifting (original scale)	235	5.0	1.9	0.0–6.0	0–6	−1.90	2.10
Attentional set shifting (binary)	235	0.7	0.5	0.0–1.0	0–1	NA	NA
Working memory (ordinal)	236	1.9	0.7	1.0–3.0	1–5	0.16	−0.85
Observer rating							
Task engagement (original scale)	234	3.5	0.6	1.5–4.0	1–4	−1.43	1.41
Task engagement (ln)	234	1.0	0.3	0.1–1.4	0–1.4	−0.89	−0.11
Teacher rating							
K learning engagement	211	4.7	1.2	1.1–6.0	1–6	−0.75	−0.22
K academic performance	207	3.2	1.3	1.0–5.0	1–5	−0.12	−1.06
G1 academic performance	194	3.1	1.3	1.0–5.0	1–5	−0.14	−1.03

Note. *N* = Sample Size; *M* = Mean; *SD* = Standard Deviation; K = Kindergarten; G1 = Grade 1; ln = natural logarithm.

Cognitive assessment. In the winter of the kindergarten school year and the spring of the first-grade year, trained research assistants administered a battery of cognitive assessments to participating children, including assessments of their expressive vocabulary (in kindergarten only) and executive functioning (in all years). Only the kindergarten cognitive assessments are used in the present analyses. The instruments used and the measures that were derived from these instruments are described below.

Expressive vocabulary. Children's expressive vocabulary was assessed in kindergarten using the Expressive One-Word Picture Vocabulary Test (Brownell, 2000). In this assessment, children were shown pictures of objects or actions and were asked to provide the word that names each picture. Standard scores were assigned based on national norms provided by the test developer (Brownell, 2000), with scores in the present sample ranging from 55 to 122 (M = 83.9, SD = 12.3). Expressive vocabulary scores were missing for three children (1.3% of the analysis sample) who did not complete this task.

Working memory. Children's working memory was assessed in kindergarten with the Backward Word Span task (Davis & Pratt, 1995), in which children were asked to repeat back verbally presented strings of words in reverse order. Children's performance on the task was coded as the maximum number of words they correctly recalled in reverse order. Children who did not pass a two-word practice trial after two attempts were assigned a score of 1. In the present sample, scores ranged from 1 to 3 (M = 1.9, SD = 0.7). Working memory scores were missing for three children (1.3% of the sample) who did not complete this task.

Inhibitory control. Children's inhibitory control was assessed in kindergarten with the Peg Tapping task (Diamond & Taylor, 1996), in which children were asked to tap a pencil once when the research assistant tapped twice and to tap twice when the research assistant tapped once. Prior to beginning the task, all children completed at least one practice trial for each rule with coaching from the research assistant; a maximum of six practice trials were administered prior to beginning the testing phase. The testing phase consisted of 16 trials with the number of taps varied in pseudorandom order. Scores were coded as the number of correct trials, from 0 to 16. Although the full range of possible scores was observed in the present sample, children's scores were strongly negatively skewed due to ceiling effects (see Table 1). Therefore, scores on this task were dichotomized as *fewer than two incorrect responses* (64.5% of the analysis sample) versus *two or more incorrect responses* (35.5% of the analysis sample). Inhibitory control scores were missing for eight children (3.3% of the analysis sample) who did not complete this task.

Attentional set shifting. Children's attentional set shifting was assessed in kindergarten with the Dimensional Change Card Sort task (Frye, Zelazo, & Palfai, 1995). In this task, children were instructed to sort cards along one dimension (either shape or color). After successfully sorting cards along this dimension on six consecutive trials, children were instructed to sort the cards along the other dimension. During the testing phase, the research assistant reviewed the sorting rules prior to each trial but did not provide feedback on the child's performance. Scores were coded as the number of correct trials following the switch in sorting dimension, from 0 to 6. Thus, scores reflect the child's ability to switch the focus of their attention between aspects of the same object and to inhibit the prepotent impulse to continue sorting according to the first dimension. Although the full range of possible scores was observed in the present sample, 71.9% of children performed correctly on all six postswitch trials. Therefore, scores on this task were dichotomized as no postswitch errors (71.9%) versus any postswitch errors (28.1%). Attentional set shifting scores were missing for four children (1.7% of the analysis sample) who did not complete this task.

Observer rating

Task engagement. At the end of the cognitive assessment, research assistants retrospectively rated children's overall level of self-regulated, positive engagement with the assessment tasks using a 13-item adaptation of the Leiter-R Assessor Report (Roid & Miller, 1997; Smith-Donald, Raver, Hayes, & Richardson, 2007).

Items included "pays attention to instructions and demonstrations," "sustains concentration," "willing to try repetitive tasks," and "modulates and regulates arousal level in self." Each item was rated on a 4-point scale with item-specific descriptive anchors. Higher scores indicated more positive task engagement. Summary scores were calculated as the mean of all item scores and exhibited high internal reliability in the current sample ($\alpha = .92$). This measure exhibited moderate negative skew and was therefore subjected to natural log transformation (see Table 1 for raw and transformed univariate statistics). Task engagement scores were missing for five children, or 2% of the analysis sample.

Neurophysiological assessment. In the fall of participating children's kindergarten school year, psychophysiological assessments were conducted by trained research assistants in a mobile research laboratory. Psychophysiological equipment was installed into a recreational vehicle (RV) and driven to each school to conduct the assessments. This maximized consistency of the testing environment across the school sites while minimizing burden on parents. The psychophysiological assessments took place during the school day and lasted approximately 45 minutes. These assessments were conducted on all but nine children in their kindergarten year. During the assessment, both EEG and autonomic physiological data were collected during a go/no-go task followed by an emotion induction task. Only the P3b event-related potential during the go/no-go task will be addressed here. Additional detail on the psychophysiological assessment procedures can be found in Gatzke-Kopp et al. (2012).

Go/No-Go task. The task used for the present analyses was patterned after a go/no-go paradigm first reported by Lewis, Lamm, Segalowitz, Stieben, and Zelazo (2006) and Stieben et al. (2007). Stimuli consisted of a series of cartoon character images that were designed to be appealing and appropriate for children in kindergarten. RAs followed a scripted protocol and used pictures of the stimuli to explain the task instructions to the participating children. The stimuli were described as "alien critters" that the participants should "zap" by pressing the response box button whenever one appeared (the *go* condition; 68% of all trials). The children were then instructed that they were not allowed to zap the same character twice in a row, so they should resist hitting the button if the same critter that they had just seen appeared again (the *no-go* condition; 32% of all trials). The task consisted of a total of 330 trials (225 go trials and 105 no-go trials) and lasted approximately 12 minutes. A total of 45 different character stimuli were used; thus, across the 225 go trials, each specific stimulus occurred with a frequency of roughly 2%.

Children were informed that during the game they would be earning points for correct answers and losing points for mistakes, which were signaled by a large red box appearing around the stimuli. Children were also told that they would win a prize if they were able to earn enough points. No specific criterion of point value was associated with earning a prize so children would be motivated to maximize points. Prizes were shown to the child ahead of time and consisted of goody bags filled with small toys, stickers, and other items. During the task, children were provided with visual feedback approximately every 10 trials informing them of their progress in earning points.

After the task was explained, children engaged in a practice session and the program calculated the child's error rate on no-go trials during that session. In order to assess error-related negativity (data not presented here), error rates were targeted to be between 40 and 60% on no-go trials. Participants whose no-go error rate exceeded this window were administered the practice session again with a longer interstimulus interval and additional coaching as needed. Participants who did not meet the minimum no-go error level were administered the practice session again with a shorter interstimulus interval to increase the challenge. Initial stimulus presentation rates for the testing phase matched those used for the last practice session, and throughout the testing phase a dynamic algorithm adjusted the interstimulus intervals in order to attempt to keep no-go error rates in the targeted window. If no-go error rates exceeded 60%, trial presentation rate was automatically slowed, whereas if error rates fell below 40% trial presentation rate was speeded.

The task was administered in three blocks, with the algorithm for awarding points differing across blocks in order to experimentally manipulate the affective context. In the first block, the algorithm awarding points strongly favored correct responses and weakly punished incorrect responses, thus resulting in a rapid accumulation of points. In the second block, this algorithm was reversed, resulting in a loss of points regardless of equivalent performance, thus representing a frustrative condition. The final block employed the same high-reward algorithm as the first block and all participants ended the game with enough points to win the prize. Because the research question of interest is not related to the effects of the affective manipulation, analyses were based on the average P3b amplitude collapsed across blocks in order to maximize the number of available trials. Notably, results did not differ when P3b amplitude was calculated across the reward blocks only, and difference scores in P3b amplitude from the reward to frustration blocks were not correlated with the variables of interest in this study.

Mean response time on correct go trials and error rates on go and no-go trials were analyzed as behavioral measures of performance on the go/no-go task. It is important to note that, due to the adaptive algorithm governing stimulus presentation rates, these measures should not be directly compared to measures of performance on nonadaptive go/no-go tasks. Go trials were coded as correct and no-go trials as incorrect if a button-press response was registered between 100 and 1,000 ms following stimulus onset. The mean response time to correct go trials was 510 ms (range = 332–680 ms). On average, children failed to respond within the designated time window on 31% of go trials (range = 2–77%), and they incorrectly responded with a button press during this time window to 39% of no-go trials (range = 3–96%). The wide ranges of children's response times and error rates indicate that performance differences were apparent despite the adaptive algorithm for stimulus presentation rates. In particular, 6.5% of the sample exceeded the maximum targeted no-go error rate of 60%, which could indicate a lack of engagement with the task demands. However, excluding these individuals from the analysis sample did not alter the results, and therefore they are included in all analyses to maximize sample size and to capture the full range of performance levels present in this sample.

EEG data acquisition. EEG was recorded using an extended 10–20 montage with a 32-channel elastic stretch BioSemi headcap with the Active Two BioSemi system (BioSemi, Amsterdam, Netherlands). Two additional electrodes were placed on the left and right mastoids, and four additional facial electrodes were used to measure eye movement. Vertical eye movements were measured from electrodes placed on the infraorbital ridges centered under the pupils of both eyes and corresponding supraorbital electrodes embedded within the cap. Horizontal eye movement was measured from electrodes placed approximately 1 cm outside the participants' right and left outer canthi. Data were recorded at 512 Hz with Actiview Software, v8.0.

P3b amplitude. EEG data were postprocessed using Brain Vision Analyzer 2.0. Voltages were rereferenced to the average of all electrode sites. EEG data were strongly affected by very low frequency power in the delta band, considered typical of young children and conceptualized as a marker of developmental immaturity (Somsen, van't Klooster, van der Molen, van Leeuwen, & Licht, 1997; Yordanova & Kolev, 2008). In order to reduce the impact of sub- and very low delta frequency noise and maximize the ability to detect individual differences in P3b amplitude in response to the experimental paradigm, we employed a 1 to 30 Hz Butterworth Zero Phase filter (see, e.g., Lewis et al., 2006). The high-pass filter at 1 Hz also served to remove very slow wave drift that was present in the data.

Correct responses on go trials, defined as those for which the child responded to the stimulus between 100 and 1,000 ms after stimulus onset, were segmented from −200 to 1,000 ms relative to stimulus onset. Trials were baseline-corrected to the mean amplitude across the 200 ms prior to stimulus onset and corrections were made for eyeblink artifacts using the Gratton and Coles algorithm, as implemented by Brain Vision Analyzer 2.0 (Gratton, Coles, & Donchin, 1983). Trials with a voltage step of more than 100 μV between sampling points or a voltage reading outside the range of −75 μV to 75 μV were marked as artifactual and were excluded from the calculation of average voltages across trials.

P3b amplitudes were calculated across correct go trials in order to assess sustained attentional engagement with the task. The task instructions require participants to actively attend to each go stimulus in order to compare it with the previous trial and encode its features for comparison with the subsequent trial, thereby requiring task-oriented attention allocation processes that are expected to elicit a P3b component (Polich, 2007). It should be noted that, unlike target stimuli in a classic perceptual oddball task, the target stimuli in the present task did not possess objectively salient (i.e., unexpected) characteristics that would be likely to automatically capture attention and reliably elicit a large P3b response across all participants in each trial. Therefore, a high mean P3b amplitude averaged across all go trials in this task would be dependent upon adequate sustained attention allocation throughout the duration of the task. Conversely, a low mean P3b amplitude could be indicative of a failure to dedicate sufficient attentional resources to the task.

Inspection of the grand-averaged waveforms for correct go trials across all electrodes revealed that, consistent with previous studies (C. C. Duncan et al., 2009; Polich, 2007), voltage increases in the time window associated with the P3b appeared maximal at the midline parietal electrode Pz. The grand-average waveform at the Pz electrode following correct go trials for children in the current analysis sample is illustrated in Figure 1. P3b amplitude was calculated at this electrode as the mean voltage in the temporal window from 500 to 700 ms poststimulus onset, consistent with the location of the peak in the average waveform (see Figure 1) and generally consistent with the temporal characteristics of the P3b observed in previous studies using similarly aged children (Pfueller et al., 2011; Thomas & Nelson, 1996). Children produced a correct go response for an average of 150.5 trials across the full task (SD = 35.1). Following EEG artifact rejection procedures, an average of 119 (SD = 38.4) artifact-free correct go trials contributed to each child's mean P3b amplitude, with the number of trials ranging from 22 to 199. P3b mean

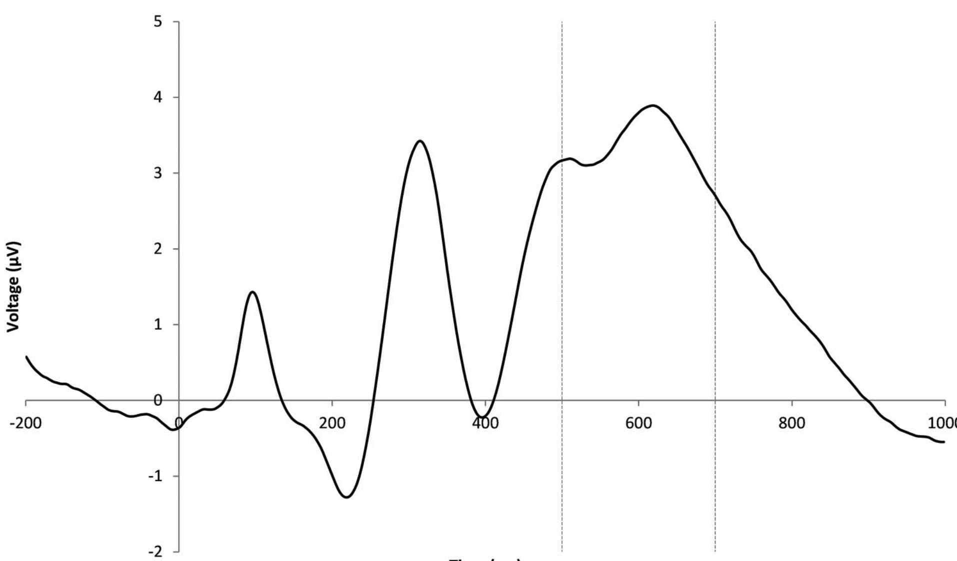

Figure 1. Grand-average waveform of correct Go trials at electrode Pz. *Note.* Time 0 marks the stimulus onset. P3b amplitude was calculated as the mean voltage at the Pz electrode in the time window from 500 to 700 ms after stimulus onset (indicated by dashed vertical lines).

amplitude was not significantly associated with the number of trials contributing to the mean, $r = -.10$, $p = .16$.

Two hundred and 11 children of the total available sample of 239 (88%) had valid P3b amplitude values. Data were missing for 28 children for the following reasons: nine did not participate in the physiological assessment, three were not assessed with EEG due to braiding of hair that prevented acceptable scalp contact, 13 did not have usable data due to equipment malfunction (e.g., broken electrodes) or RA error, and three were excluded from analyses due to values above or below 3 *SD* from the sample mean.

Results

Bivariate Statistics

Pairwise correlations among all analysis variables are presented in Table 2. Children with higher mean P3b amplitudes to go stimuli tended to have slightly longer mean response times on correct go trials, $r = .12$, $p = .08$ and had lower error rates on no-go trials, $r = -.24$, $p < .01$ but did not differ in go trial error rates. This pattern of correlations suggests that children with higher P3b amplitudes are completing the task more carefully, with less impulsive behavioral responding. Children with higher mean P3b amplitudes also tended to have a larger expressive vocabulary, $r = .26$, $p < .01$ and were more likely to perform well on assessments of inhibitory control (biserial $r = .32$, $p < .01$) and attentional set shifting (biserial $r = .21$, $p = .02$), but did not differ in working memory (polyserial $r = .10$, $p = .17$).

Children's P3b amplitudes in kindergarten were significantly correlated with concurrent teacher-rated learning engagement, $r = .18$, $p = .01$ and observer-rated task engagement, $r = .23$, $p < .01$, as well as with teacher-rated academic performance a year later, $r = .16$, $p = .04$. However, P3b amplitude was not significantly correlated with concurrent academic performance in kindergarten, $r = .07$, $p = .33$. These correlations provide support for the first hypothesis, that higher P3b amplitude would be associated with more positive learning-related behaviors and higher academic performance, although the association with academic performance only emerged as statistically significant after a 1-year lag.

The kindergarten teacher-rated learning engagement and observer-rated task engagement measures demonstrated high agreement between independent raters observing the children under different conditions, $r = .57$, $p < .01$, suggesting that these measures may reflect a common underlying construct related to children's ability to productively engage in academic contexts. Additionally, both teacher-rated learning engagement and observer-rated task engagement measures exhibited moderately large correlations with concurrent teacher-rated academic performance in kindergarten (rs = .65 and .43, respectively; ps $< .01$) as well as with first-grade teacher-rated academic performance (rs = .55 and .44, respectively; ps $< .01$). Although provided by different teachers, there was a high correlation between teacher ratings of kindergarten and first-grade academic performance, $r = .67$, $p < .01$. Additionally, P3b amplitude was not correlated with the child's age in kindergarten, $r = -.09$, $p = .17$ and did not differ significantly according to the child's gender ($t_{209} = 0.92$, $p = .36$) or race/ethnicity (African American, Hispanic, or Caucasian/Asian: $F_{2,\ 208} = 0.09$, $p = .91$).

Mediation Models

Structural equation models were constructed to test the hypothesis that kindergarten learning-related behaviors would mediate an indirect effect of kindergarten P3b amplitude on growth in academic performance between kindergarten and first grade. By predicting residualized gain in academic performance rather than absolute academic performance levels in first grade, these models yield a more rigorous directional test of the relationship between kindergarten learning-related behaviors and subsequent academic performance. These analyses were conducted first using separate models for teacher-rated learning engagement and observer-rated task engagement as independent measures of children's learning-related behaviors. Subsequently, these two measures were modeled as indicators of a common, latent mediator, in order to reduce bias in the parameter estimates due to measurement error in the mediator.

All structural equation models were conducted in LISREL 9.10 using full-information maximum-likelihood (FIML) estimation. FIML estimation takes advantage of all available information provided by the full sample of 239 children, with each case providing information regarding the variables that are not missing for that case. The statistical significance of indirect effects was calculated using the Sobel z test as implemented by LISREL 9.10.

Manifest-variable mediation models. The models testing teacher-rated learning engagement and observer-rated task engagement separately as mediators of the association between kindergarten P3b amplitude and first-grade academic performance are illustrated in Figure 2. The results of these models were strikingly similar. Both teacher and observer ratings of children's learning-related behaviors in kindergarten significantly predicted first-grade academic performance even after controlling for prior-year academic performance with a standardized effect size (β) of roughly 0.2. Sobel's z tests revealed that the indirect of effect of kindergarten P3b amplitude on residualized gain in academic performance through kindergarten teacher-rated learning engagement was significant at the trend level ($\beta = .02$, $z = 1.83$, $p = .07$), and the indirect effect through kindergarten observer-rated task engagement was significant at $p < .05$ ($\beta = .02$, $z = 2.18$, $p = .03$).

Latent mediator model. Since teacher-rated learning engagement and observer-rated task engagement were independent but strongly correlated ($r = 0.57$) ratings of a similar set of behaviors (e.g., self-control and persistence) that are important for school success, a measurement model was constructed in which these variables functioned as indicators of a common, latent variable which we refer to here as *learning-related behaviors*. To test the fit of this model to the data, teacher-rated learning engagement and observer-rated task engagement were standardized and constrained to load on a single latent variable with independent measurement errors. To allow for model identification, the covariance of observer-rated task engagement with the latent variable was fixed to 1, and all other manifest variables (kindergarten P3b amplitude, kindergarten academic performance, and first-grade academic performance) were allowed to freely covary with one another and with this latent variable. This model fit the data well ($\chi^2 = 5.88$, $df = 3$, $p = .12$; RMSEA = .06).

The latent learning-related behaviors construct was then tested as a mediator of the indirect effect of children's P3b amplitude in kindergarten on residualized gain in teacher-rated academic per-

Table 2
Pairwise Correlations Among All Analysis Variables

		Go/No-Go task			Cognitive assessment				Observer rating	Teacher rating	
	P3b amplitude	1	2	3	4	5	6	7	8	9	10
Go/No-Go task											
1. Mean response time	0.12† (211)	—									
2. Go error rate	0.01 (211)	0.36** (217)	—								
3. No-Go error rate	−0.24** (211)	−0.15* (217)	−0.39** (217)	—							
Cognitive assessment											
4. Expressive vocabulary	0.26** (210)	−0.15* (216)	−0.14* (216)	−0.19** (216)	—						
5. Inhibitory control (binary)	0.32** (206)	−0.20* (212)	−0.05 (212)	−0.23** (212)	0.18* (231)	—					
6. Attentional set shifting (binary)	0.21* (210)	−0.22* (216)	−0.01 (216)	−0.07 (216)	0.29** (235)	0.15 (231)	—				
7. Working memory (ordinal)	0.10 (210)	−0.17* (216)	−0.12 (216)	−0.13† (216)	0.28** (236)	0.33** (231)	0.36** (235)	—			
Observer rating											
8. Task engagement (ln)	0.23** (208)	−0.14* (214)	−0.07 (214)	−0.15* (214)	0.26** (234)	0.46** (229)	0.12 (233)	0.34** (234)	—		
Teacher rating											
9. K learning engagement	0.18* (187)	−0.12† (192)	−0.08 (192)	−0.10 (192)	0.33** (208)	0.31** (203)	0.21* (207)	0.42** (208)	0.57** (206)	—	
10. K academic performance	0.07 (185)	−0.31** (190)	−0.17* (190)	−0.10 (190)	0.48** (204)	0.35** (199)	0.24** (203)	0.51** (204)	0.43** (202)	0.65** (206)	—
11. G1 academic performance	0.16* (174)	−0.27** (180)	−0.14† (180)	−0.10 (180)	0.41** (192)	0.38** (188)	0.10 (192)	0.44** (192)	0.44** (190)	0.55** (170)	0.67** (167)

Note. Pearson correlation coefficients are reported for continuous-continuous variable pairs, polyserial correlation coefficients for ordinal-continuous or binary-continuous variable pairs, polychoric correlation coefficients for ordinal-binary variable pairs, and tetrachoric correlation coefficients for binary-binary variable pairs. Pairwise sample sizes are reported in parentheses. K = Kindergarten; G1 = Grade 1; ln = natural logarithm.
† $p < .10$. * $p < .05$. ** $p < .01$.

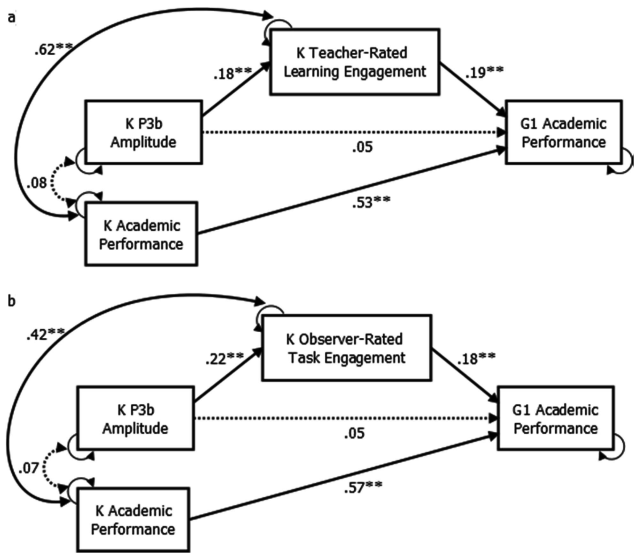

Figure 2. Manifest-variable mediation models. *Note.* Standardized path coefficients are presented for a) teacher-rated learning engagement and b) observer-rated task engagement manifest-variable mediation models. Dotted lines represent parameter estimates that are not significant at $p < .05$. K = Kindergarten; G1 = Grade 1. † $p < .10$; * $p < .05$; ** $p < .01$.

formance between kindergarten and first grade. The results of the latent mediator model are illustrated in Figure 3. As expected, this model provided the same pattern of findings as was observed in the manifest-mediator models. Specifically, kindergarten P3b amplitude was significantly associated with kindergarten learning-related behaviors, which in turn predicted first-grade academic performance controlling for prior-year academic performance. This latent mediator model yielded a larger standardized effect size for the path from the mediator (learning-related behaviors) to the outcome (first-grade academic performance) than was observed in either of the manifest-variable mediation models ($\beta = .31$ vs. .19 and .18), presumably reflecting a reduction in measurement error for the mediator. The indirect effect of P3b amplitude on residualized gain in 1st-grade academic performance through the latent kindergarten learning-related behaviors construct was statistically significant according to the Sobel *z* test ($\beta = .08$, $z = 2.10$, $p = .04$).

Testing for moderation by gender. Multiple-group structural equation analyses were conducted to test whether the parameter estimates for the latent mediator model varied between male ($n = 153$) and female ($n = 86$) children. This test was conducted by comparing the fit of a multiple-group model in which all parameter estimates were allowed to vary freely across gender groups (the *unconstrained* model) to that of a model in which all parameter estimates were constrained to be equivalent across gender groups (the *fully constrained* model). The unconstrained multiple-group model did not provide a significantly better fit for the data than did the fully constrained model ($\Delta\chi^2 = 11.9$, $\Delta df = 13$, $p = .53$). Moreover, the fully constrained model exhibited good absolute fit to the data ($\chi^2 = 16.1$, $df = 17$, $p = .52$, RMSEA = 0.00). These results suggest that the parameter estimates for the latent mediator model do not vary significantly between male and female children in this sample.

Testing for moderation by aggression classification at school entry. Since children in the present study had either relatively high ($n = 107$) or low ($n = 132$) levels of teacher-rated aggressive/oppositional behaviors at the beginning of their kindergarten school year, it is important to examine whether the associations reported for the full sample hold equally for the children in these high and low early aggression groups. This test was conducted according to the same multiple-group structural equation modeling procedures that were described above for the gender moderation analyses. The unconstrained model did not provide a better fit to the data than did the fully constrained model ($\Delta\chi^2 = 8.5$, $\Delta df = 13$, $p = .81$). Additionally, the fully constrained multiple-group model exhibited good absolute fit to the data ($\chi^2 = 13.8$, $df = 17$, $p = .68$, $RMSEA = 0.00$). Thus, the parameter estimates for the latent mediator model do not vary significantly according to whether the child was rated as high or low on aggressive/oppositional behaviors at school entry.

Unique predictive value of P3b amplitude. In order to assess the degree to which P3b amplitude accounts for unique variance in children's learning-related behaviors beyond that which can be accounted for by behavioral assessments of cognitive function, we reexamined the latent mediator model controlling in turn for each of the cognitive–behavioral performance measures from the go/no-go task (mean response time, go error rate, and no-go error rate) and from the cognitive assessment (expressive vocabulary, inhibitory control, attentional set shifting, and working memory) administered in kindergarten. In each of these models, the cognitive–behavioral measure was entered as a covariate predicting both the learning-related behaviors latent construct and first-grade academic performance. Thus, the paths from P3b amplitude to learning-related behaviors and from learning-related behaviors to first-grade academic performance were both adjusted for the cognitive–behavioral covariate. Table 3 provides the unadjusted and adjusted parameter estimates for the paths from P3b amplitude to learning-related behaviors (path a') and from learning-related behaviors to first-grade academic performance (path b'), as well as for the indirect effect from P3b amplitude to first-grade academic performance via learning-related behaviors (paths $a' \cdot b'$).

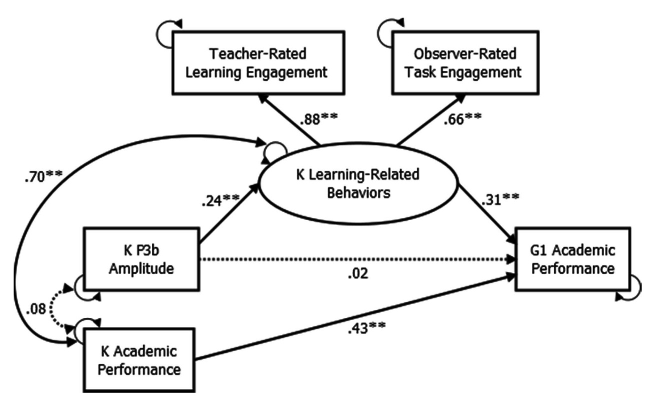

Figure 3. Latent mediator model. *Note.* Standardized path coefficients are presented. Dotted lines represent parameter estimates that are not significant at $p < .05$. K = Kindergarten; G1 = Grade 1. † $p < .10$; * $p < .05$; ** $p < .01$.

Table 3
Parameter Estimates for the Latent Mediator Model, Unadjusted and Controlling for Various Kindergarten Cognitive–Behavioral Measures

	a′				b′				a′ · b′			
Cognitive–behavioral covariate	estimate	*SE*	*t*	β	estimate	*SE*	*t*	β	estimate	*SE*	*t*	β
Unadjusted	0.07	0.02	3.06	0.24**	0.59	0.21	2.83	0.31**	0.04	0.02	2.10	0.08*
Go/No-Go task												
Mean response time	0.08	0.02	3.32	0.26**	0.62	0.21	3.00	0.32**	0.05	0.02	2.25	0.08*
Go error rate	0.07	0.02	3.08	0.24**	0.60	0.21	2.88	0.31**	0.04	0.02	2.13	0.08*
No-Go error rate	0.07	0.02	2.81	0.23**	0.60	0.21	2.85	0.31**	0.04	0.02	2.02	0.07*
Cognitive assessment												
Expressive vocabulary	0.04	0.02	1.93	0.15*	0.60	0.21	2.93	0.31**	0.03	0.02	1.62	0.05
Inhibitory control (binary)	0.05	0.02	2.10	0.16*	0.63	0.23	2.78	0.34**	0.03	0.02	1.69	0.06†
Attentional set shifting (binary)	0.06	0.02	2.79	0.22**	0.60	0.21	2.89	0.31**	0.04	0.02	2.02	0.07*
Working memory (ordinal)	0.06	0.02	2.82	0.21**	0.59	0.21	2.79	0.31**	0.04	0.02	2.00	0.06*

Note. a′ = direct path from P3b amplitude to learning-related behaviors; b′ = direct path from learning-related behaviors to first-grade academic performance; a′ · b′ = indirect path from P3b amplitude to first-grade academic performance via learning-related behaviors; estimate = unstandardized parameter estimate; *SE* = standard error; *t* = *t*-statistic; β = standardized parameter estimate.
† $p < .10$. * $p < .05$. ** $p < .01$.

P3b amplitude remained a significant predictor of the latent learning-related behaviors construct when each of these cognitive–behavioral measures was included as a covariate, with standardized effect sizes ranging from 0.15 to 0.26. Thus, P3b amplitude accounted for unique variance in children's learning-related behaviors that could not be fully explained by any one of these cognitive–behavioral measures. Likewise, the association between kindergarten learning-related behaviors and residualized gain in academic performance remained significant, with no meaningful changes in the magnitude of this association, when each of these cognitive–behavioral measures was covaried. Finally, the magnitude of the indirect effect of P3b amplitude on residualized gain in academic performance via kindergarten learning-related behaviors was similar in all tested models (with standardized effect sizes ranging from 0.05 to 0.08), although the statistical significance of this estimate fell below the conventional .05 threshold in two models: those covarying children's expressive vocabulary ($\beta = 0.05$, $t = 1.62$, $p = .11$) and their inhibitory control ($\beta = 0.06$, $t = 1.69$, $p = .09$).

Discussion

The purpose of this study was to provide insight into how individual differences in a neurophysiological measure of controlled attention allocation—amplitude of the P3b event-related potential to task-relevant stimuli—are associated with learning-related behaviors and academic performance within a low-SES sample of early school-age children. Results revealed that kindergarten children who had higher mean P3b amplitudes to target stimuli in a go/no-go task tended to have higher levels of concurrent learning-related behaviors, which in turn predicted growth in academic performance between kindergarten and first grade. These findings replicated across different raters of children's learning-related behaviors (i.e., teachers and trained observers) who were basing their assessments on different observational settings (i.e., the day-to-day classroom environment and a direct cognitive–behavioral assessment). Since P3b amplitude to task-relevant stimuli indexes controlled attention allocation to the task, our findings provide further support for the importance of attention skills for successful school adjustment and academic success (G. J. Duncan et al., 2007; Lan et al., 2011). Children with high mean P3b amplitudes may function better in the classroom because they are better able to maintain focus on the task at hand—whether this is listening to the teacher or working on an assignment—while ignoring both external and internal distracters.

The present study expands on a small number of prior studies that have also reported positive associations between P3b amplitude and academic success (Harmon-Jones et al., 1997; Hillman et al., 2012). These prior studies have observed cross-sectional associations between higher P3b amplitudes and higher concurrent reading achievement in pre- and early adolescence. By assessing this association in younger children, our findings provide evidence that individual differences in P3b amplitude measured as early as kindergarten relate to academic performance in the following school year. Importantly, this study also provides evidence that P3b amplitude indirectly predicts growth in academic performance from kindergarten to 1st grade through its association with children's learning-related behaviors in kindergarten. Notably, the association between P3b amplitude and children's learning-related behaviors remained significant even when controlling for children's expressive vocabulary at school entry, suggesting that this association is not solely attributable to a relation between P3b amplitude and early language skills.

Studies examining child-level factors that predict successful transitions into the school environment often focus on inhibitory control as laying the groundwork for school success (Blair & Diamond, 2008). In our sample, P3b amplitude was positively associated with performance on two different behavioral assessments of inhibitory control as well as with a measure of attentional set shifting (which, as assessed in this study, also requires cognitive inhibition of the prior task set). These findings are consistent with prior research showing that higher P3b amplitudes are associated with lower impulsivity and distractibility in older children and adults (Boucher et al., 2010; Harmon-Jones et al., 1997; Määttä et al., 2005; Portin et al., 2000; Russo et al., 2008), while extending this pattern of findings to early school-age children. Importantly, however, in the present study P3b amplitude predicted unique variance in children's learning-related behaviors that was not accounted for by any one of these behavioral measures of inhibitory control or attentional set shifting. This suggests that those aspects of controlled attention allocation that are indexed by P3b amplitude in the present study may contribute to learning-related behaviors at school entry above and beyond the contribution of other self-regulatory skills such as inhibitory control or attentional set shifting. More generally, this finding highlights how neurophysiological measures can provide information about neurocognitive processes underlying successful school adjustment that may not be fully captured by behavioral measures of cognitive functioning.

It is interesting to note that in the present study P3b amplitude predicted teacher-rated academic performance in first grade but was not associated with concurrent academic performance in kindergarten. This finding is consistent with the hypothesis that the association between P3b amplitude and later academic performance is mediated by classroom behavior rather than arising from an unmeasured confounding factor such as IQ. Thus, children with poor attention skills may fall progressively farther behind their peers over time due to the accumulation of missed learning opportunities. Furthermore, children who experience academic underachievement may be at increased risk of developing externalizing behaviors and substance dependence (Crum, Ensminger, Ro, & McCord, 1998; Hinshaw, 1992). Thus, academic difficulties represent an understudied potential pathway that could contribute to the frequently observed association between P3b amplitude and externalizing psychopathology in adolescents (Patrick et al., 2006). These observations highlight the potential importance of early intervention for children with low P3b amplitudes to task-relevant stimuli, which may be indicative of attention problems. Further studies are required to evaluate longer-term trajectories of learning-related behaviors and academic and behavioral risk associated with low P3b amplitudes.

Implications for Science, Practice and Policy

The demonstration of an association between P3b amplitude and learning-related behaviors specifically in the early school years is particularly relevant to education practice and policy, as the rapid cognitive development during this age suggests that it may be a valuable time period on which to focus preventive intervention efforts. Assessing neurophysiological indices of attention allocation prior to and following these interventions may provide useful insight into the mechanisms of change. For example, a multicomponent attention training intervention for low-SES preschool children has been shown to yield improvements in a neurophysiological measure of early selective attention (in the time range of the N1 event-related potential) which are accompanied by improvements in language skills and IQ (Neville et al., 2013). There is also intriguing evidence that mindfulness meditation (which involves self-regulation of attention toward the present experience) increases P3b amplitude to target stimuli immediately following the meditation exercise in experienced adult practitioners (Delgado-Pastor, Perakakis, Subramanya, Telles, & Vila, 2013). Thus it seems probable that the neural processes contributing to P3b amplitude to target stimuli are sensitive to cognitive–behavioral interventions. Results from the present study suggest that future research should examine (a) whether early interventions targeting children's attention skills yield increases in P3b amplitudes to target stimuli and, if so, whether these P3b amplitude increases are associated with improvements in learning-related behaviors in school, and (b) whether children with lower P3b amplitudes to target stimuli are particularly likely to show improvements in learning-related behaviors from early interventions targeting attention skills. These approaches could advance the state of early intervention science by focusing on key mechanisms, such as attention control, that support successful school adjustment.

Furthermore, the children included in the present study were drawn from a population of great importance for social policy. These children were exposed to high levels of socioeconomic adversity and were predominantly members of ethnic minority groups that are frequently underrepresented in neurophysiological and psychological studies. Support for a relationship between P3b amplitude to task-relevant stimuli, as a neurophysiological measure of controlled attention allocation, and these children's learning-related behaviors and academic performance provides insight into one factor that may contribute to risk and resilience within socially disadvantaged contexts. Whether these results can be considered to generalize to populations with higher socioeconomic status or majority race/ethnicity, however, awaits further scientific study.

Limitations

The results and implications of these analyses must be interpreted with certain limitations in mind. One important limitation is that, since kindergarten P3b amplitude and kindergarten learning-related behaviors are assessed more or less concurrently and without controls for prior levels of learning-related behaviors, the current analyses do not provide statistical evidence for causal effects of P3b amplitude on learning-related behaviors. It is possible that unmeasured variables, such as a disorganized home environment or parent psychopathology, influence both of these measures. Such potential confounding variables could not be included in the present analyses because data on the home environment and parental mental health were not collected for all participants. However, the present analyses did show that P3b amplitude is associated with children's learning-related behaviors even when a variety of cognitive–behavioral assessments are controlled.

Another limitation of the present analyses was the use of teacher-reported academic performance as the outcome variable rather than a more objective measure such as standardized test scores. These analyses should be replicated with an objective academic assessment before they can be assumed to reflect the effects of P3b amplitude and learning-related behaviors on children's true academic competence in the early school years. However, prior studies have observed associations between P3b amplitude and standardized reading test scores among older children (Harmon-Jones et al., 1997; Hillman et al., 2012).

Additionally, it is important to note that the precise neurocognitive processes indexed by P3b amplitude are not entirely clear. In particular, it has been suggested that the P3b reflects working memory updating operations, a more specific neurocognitive process than attention allocation (Donchin, 1981). However, there is an abundance of evidence that experimentally manipulating conscious attention allocation to a stimulus manipulates the P3b amplitude following this stimulus, whether or not the task specifically requires working memory updating operations (see Picton, 1992). Since the encoding of information in working memory is facilitated when this information is actively attended, working memory updating is strongly influenced by more general attention allocation processes that are indexed by P3b amplitude. Our interpretation of P3b amplitude as reflecting attention allocation is grounded in an empirically based theory of the neural processes responsible for generating this event-related potential that parsimoniously accounts for the expansive experimental evidence on this topic (Nieuwenhuis et al., 2005). Nonetheless, it is still possible that the individual differences in P3b amplitude observed in the current sample may reflect working memory updating processes in addition to more general attention allocation, particularly since the go/no-go task used in this study required children to continually update their working memory trace in order to detect consecutively repeated stimuli. This interpretation is weakened, however, by the lack of association between P3b amplitude and working memory performance. Further research on the neurocognitive processes captured by individual differences in P3b amplitude could help researchers develop and refine more sensitive and specific behavioral indices to identify children at risk for academic difficulties. Our results demonstrate that standard neuropsychological assessments, such as basic working memory and other executive function tasks, do not fully account for the individual differences in P3b amplitude or the association between P3b amplitude and learning outcomes.

Finally, in interpreting these findings, the reader should keep in mind that the current task involved an affective manipulation, in which children acquired points toward a prize during the first and last blocks but lost points toward this prize during the middle block. Although the findings did not differ when the middle block was excluded from the calculation of mean P3b amplitude, it is still true that P3b amplitude was measured in an affectively charged context and therefore the results cannot be assumed to generalize to P3b amplitude measures collected in affectively neutral contexts. This being said, the affective context used in the present study may be more representative of the school context, in which performance is associated with rewarding and punishing implications.

Conclusion

This study provides evidence that, among urban kindergarten children with low socioeconomic resources, higher P3b amplitudes are associated with more positive learning-related behaviors in the kindergarten classroom. Learning-related behaviors in turn mediate a positive indirect effect of P3b amplitude on gain in academic performance from kindergarten to first grade. These findings fill a significant gap in the research literature by providing support for a specific developmental process that may underlie the association between P3b amplitude and academic achievement, revealing that this association emerges as early as first grade, and demonstrating this association within a socioeconomically disadvantaged sample. Since neurophysiological research suggests that P3b amplitude reflects the activity of neural networks underlying attention allocation, these findings build on the scientific justification for interventions targeting attention skills in at-risk young children in order to promote positive learning-related behaviors and academic achievement.

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